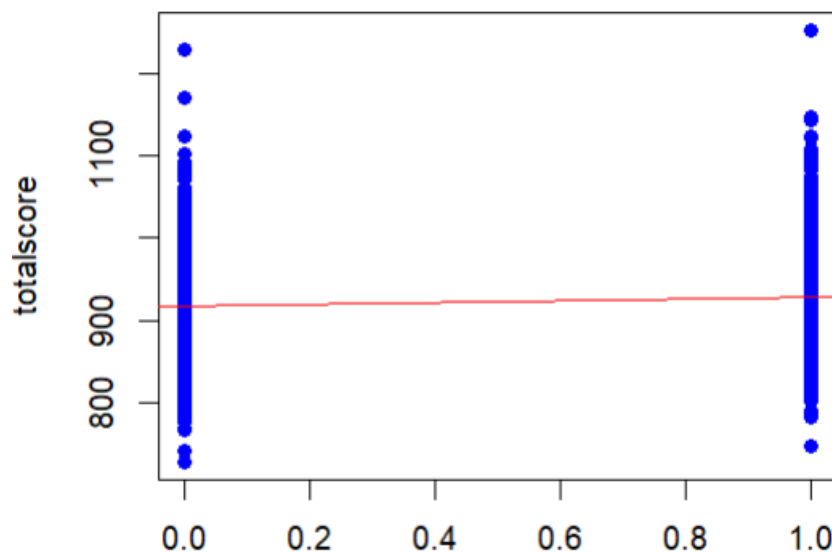


(a)

```
#(a)
idx <- star5_small$regular == 1 | star5_small$small == 1
#取出資料框中的 regular 以及 small 的變數(==1表示是該類型的資料、|代表or)
#結論idx會是一個TRUE OR FALSE的向量，TRUE表示保留該筆資料
df <- star5_small[idx, ]
#資料框格式 -> data[列, 欄]
#idx是篩選列 -> star5_small[idx, ]是只保留idx = TRUE的列(只包含regular & small且排除aide class)
mod1 <- lm(totalscore ~ small, data = df)
#totalscore ~ small指的是totalscorei = β0 + β1 * smalli + ui
intercept <- coef(mod1)[1]
#取出截距；[1]表示第一個元素
#coef(mod1)會得到截距為β0且small的係數為β1
slope <- coef(mod1)[2]
#####
plot(df$small, df$totalscore, #前面為x軸，後面為y軸
      xlab="small",
      ylab="totalscore",
      main = "Observations and Fitted line", #圖片標題

      pch = 16,col = 'blue', #pch表示實心圓點
      type = "p") #p = point 只畫點
abline(intercept,slope ,col = 'red') #abline為畫一條直線，前兩者為參數
cat(
  "b1 is:", intercept,"when small=0,E[Totalscore|small=0]=b1", "\n",
  "b2 is:", slope,"when small increase 1(class is small),totalscore increase b2", "\n",
  "since b2>0,small class has higher totalscore,but the points in the two groups are widely dispersed.", "\n",
  "so it may not Statistically significant", "\n",
  sep = ""
)
#cat()把多個文字與變數 串接(concatenate) 起來並直接輸出到 console
#""\n" 代表換行
## b1 is:916.4417,when small=0,E[Totalscore|small=0]=b1
## b2 is:12.17533,when small increase 1(class is small),totalscore increase b2
## since b2>0,small class has higher totalscore,but the points in the two groups are widely dispersed.
## so it may not Statistically significant
```

Observations and Fitted line



(b)

```
#(b)
mod2 <- lm(readscore ~ small, data = df)
#β0=regularclass 平均閱讀分數
#β1=smallclass 與 regularclass 的平均差
intercept_2 <- coef(mod2)[1]
slope_2 <- coef(mod2)[2]

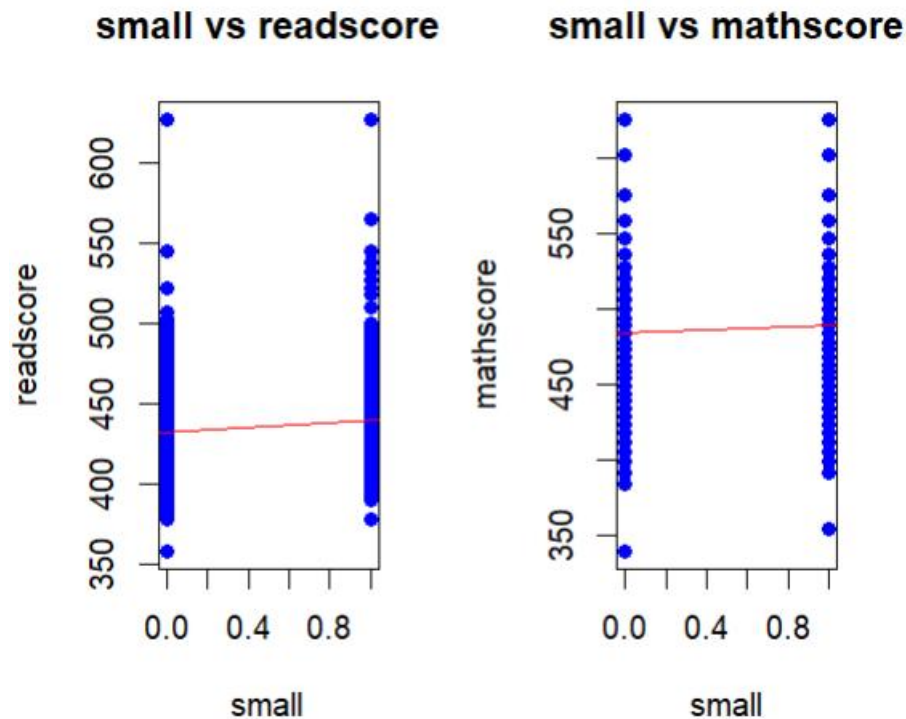
mod3 <- lm(mathscore ~ small, data = df)
intercept_3 <- coef(mod3)[1]
slope_3 <- coef(mod3)[2]
#β0=regularclass 平均是學分數
#β1=smallclass 與 regularclass 的平均差
#####
par(mfrow = c(1, 2))
#par() = 設定圖形參數
#mfrow = c(1,2)意思是
#1 row
#2 columns
#也就是兩張圖並排 => [圖一] [圖二]

plot(df$small, df$readscore, #畫閱讀成績點散圖
      xlab="small",
      ylab="readscore",
      main = "small vs readscore",
      pch = 16,col = 'blue',
      type = "p") #圖形型態(只畫點)
abline(intercept_2,slope_2 ,col = 'red')

plot(df$small, df$mathscore, #畫數學成績點散圖
      xlab="small",
      ylab="mathscore",
      main = "small vs mathscore",
      pch = 16,col = 'blue',
      type = "p")
abline(intercept_3,slope_3 ,col = 'red')

par(mfrow = c(1, 1)) # 恢復為單一圖形版面，之後再畫圖就不會出現並排圖

cat(
  "b1_read is:", intercept_2, ".b1_math is:", intercept_3, "\n",
  "b2_read is:", slope_2, ".b2_math is:", slope_3, "\n",
  "readscore is more improved than mathscore when class is small", "\n",
  "but it may not Statistically significant", "\n",
  sep = ""
)
## b1_read is:432.665.b1_math is:483.7767
## b2_read is:6.924483.b2_math is:5.250849
## readscore is more improved than mathscore when class is small
## but it may not Statistically significant
```



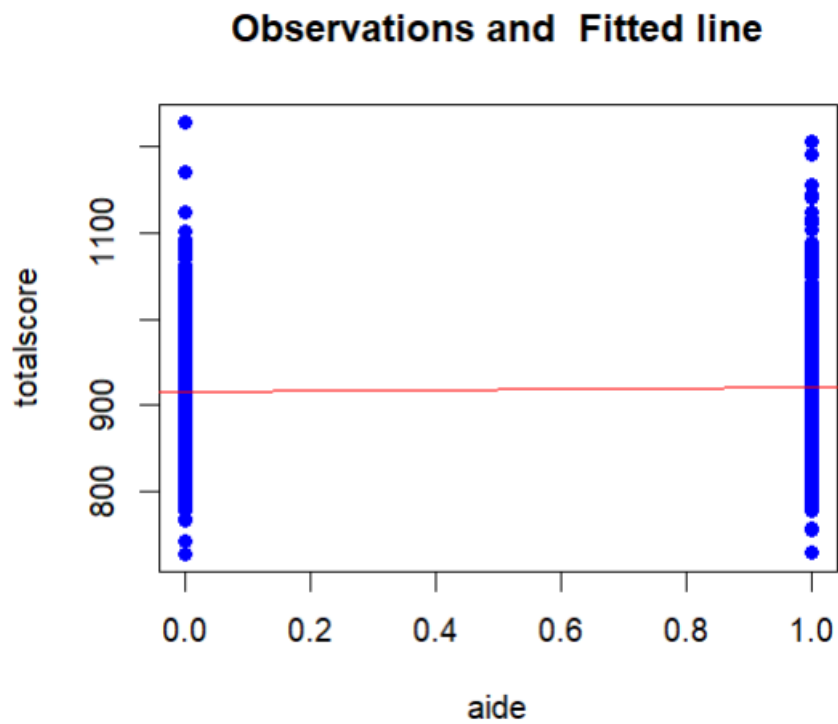
(c)

```
#(c)
idx2 <- star5_small$regular == 1 | star5_small$aide == 1

df2 <- star5_small[idx2, ]
mod4 <- lm(totalscore ~ aide, data = df2)
intercept_4 <- coef(mod4)[1]
slope_4 <- coef(mod4)[2]

#####
plot(df2$aide, df2$totalscore,
     xlab="aide",
     ylab="totalscore",
     main = "Observations and Fitted line",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_4,slope_4 ,col = 'red')

cat(
  "b1 is: ", intercept_4,"when aide=0,E[Totalscore|aide=0]=b1", "\n",
  "b2 is: ", slope_4,"when aide increase 1(class has aide),totalscore increase b2", "\n",
  "since b2>0,regular class with aide has higher totalscore,but the points in the two groups are widely dispersed.", "\n",
  "so it may not Statistically significant", "\n",
  sep = ""
)
## b1 is: 916.4417,when aide=0,E[Totalscore|aide=0]=b1
## b2 is: 4.306488,when aide increase 1(class has aide),totalscore increase b2
## since b2>0,regular class with aide has higher totalscore,but the points in the two groups are widely dispersed.
## so it may not Statistically significant
```



(d)

```
#(d)
mod5 <- lm(readscore ~ aide, data = df2)
intercept_5 <- coef(mod5)[1]
slope_5 <- coef(mod5)[2]

mod6 <- lm(mathscore ~ aide, data = df2)
intercept_6 <- coef(mod6)[1]
slope_6 <- coef(mod6)[2]

#####
par(mfrow = c(1, 2))
plot(df2$aide, df2$readscore,
     xlab="aide",
     ylab="readscore",
     main = "aide vs readscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_5,slope_5,col = 'red')
```

```

plot(df2$aide, df2$mathscore,
     xlab="aide",
     ylab="mathscore",
     main = "aide vs mathscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_6,slope_6 ,col = 'red')
par(mfrow = c(1, 1))

cat(
  "b1_read is:", intercept_5,".b1_math is:", intercept_6, "\n",
  "b2_read is:", slope_5,".b2_math is:", slope_6, "\n",
  "readscore is more improved than mathscore when class has aide", "\n",
  "but it may not Statistically significant", "\n",
  sep = ""
)
## b1_read is:432.665.b1_math is:483.7767
## b2_read is:2.871422.b2_math is:1.435066
## readscore is more improved than mathscore when class has aide
## but it may not Statistically significant

```

